SKILL ASSESSMENT TEST – 2

U23IT484 – CLOUD COMPUTING TECHNOLOGIES

NAME : SHUBHAHARINI S

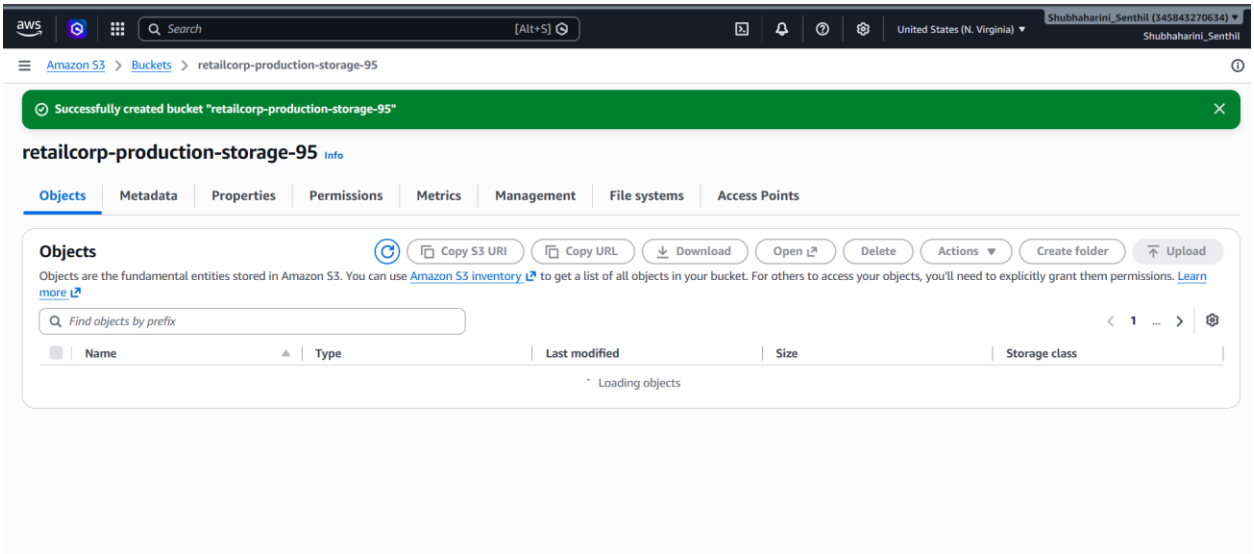
ROLL NO: 24AM095

TITLE : AWS ACTIVITY AUDITING

Using AWS CloudTrail, Amazon S3, IAM, and CloudWatch Logs implementing centralized AWS activity logging.

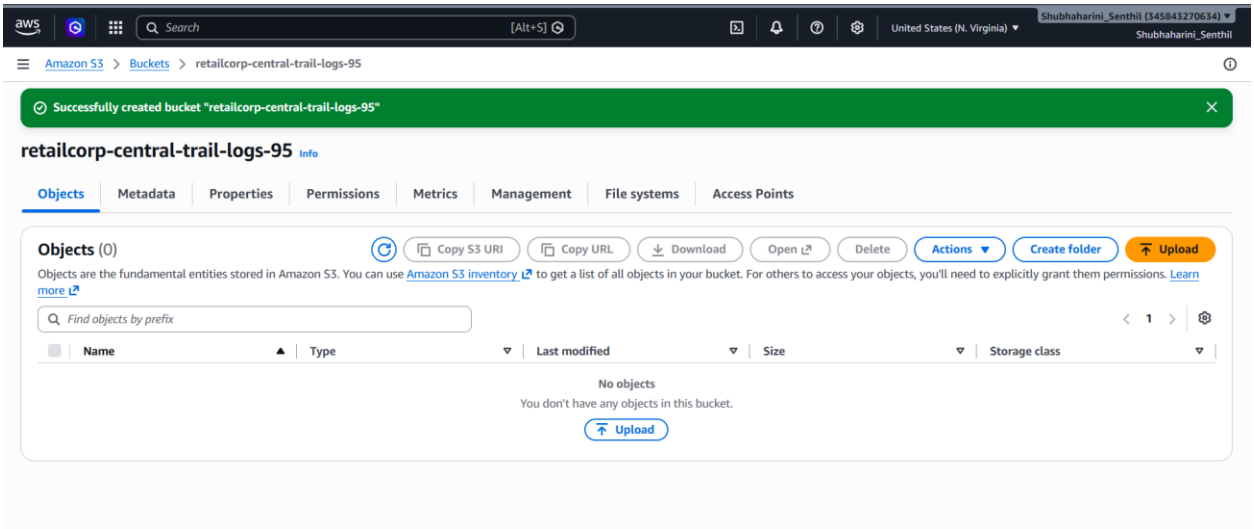
1. Creating the RetailCorp Production S3 Bucket

The primary production storage bucket, retailcorp-production-storage-95, to host critical business assets was created, which represents the operational repository subject to forensic monitoring.



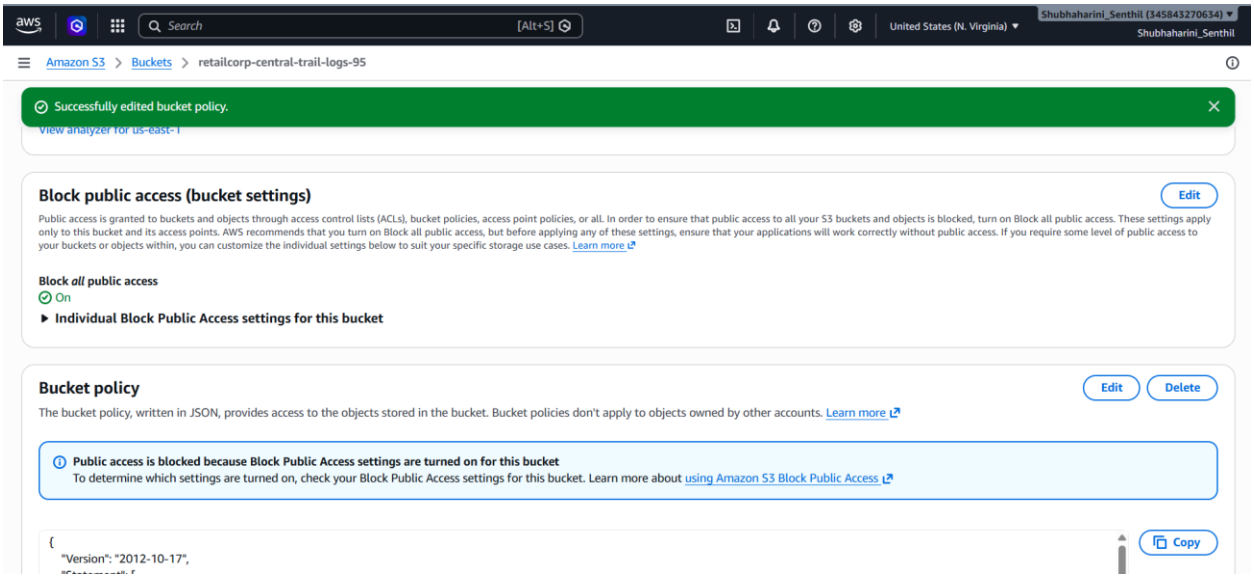
2. Creating Provision Dedicated Central Audit Log Bucket

A dedicated repository bucket, retailcorp-central-trail-logs-95, was deployed to receive centralized audit log archives from AWS CloudTrail. Storing compliance archives separately from transactional data ensures privilege isolation and protects forensic records against unauthorized tampering or accidental modification.



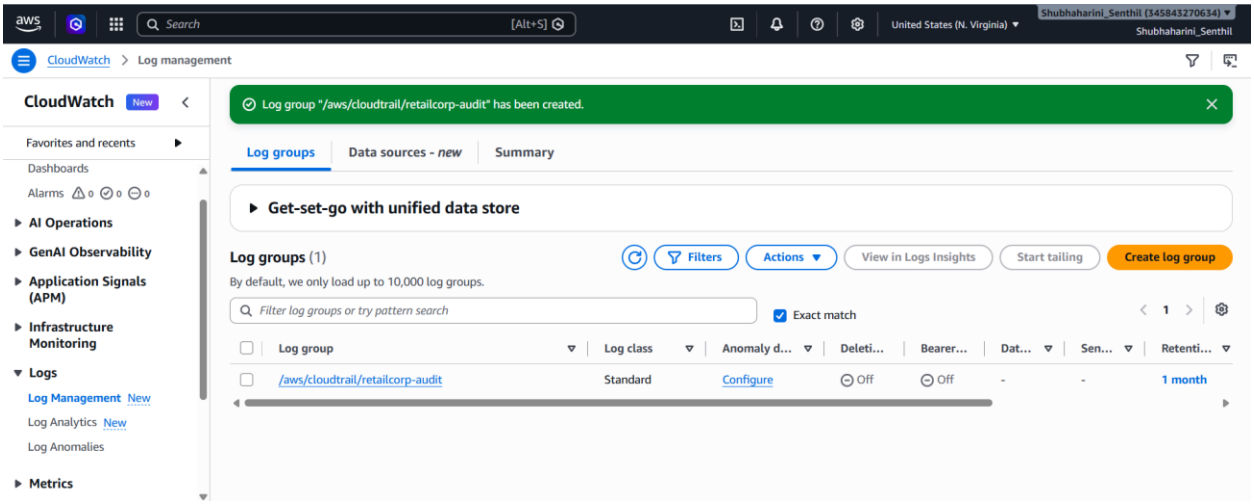
3. Configuring CloudTrail Delivery Policy on the Audit Bucket

An S3 bucket policy was attached to the central audit bucket granting the cloudtrail.amazonaws.com service principal permission to verify bucket ACLs and deliver compressed .json.gz log streams.



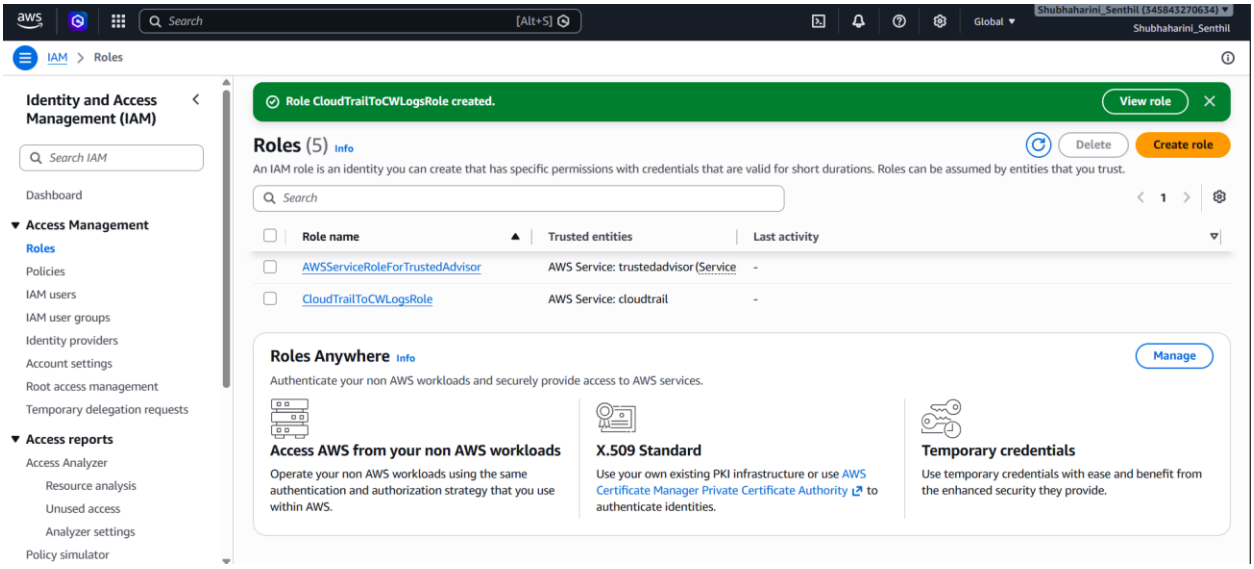
4. Creating CloudWatch Log Group for Real-Time Ingestion

Configured a dedicated CloudWatch Log Group, /aws/cloudtrail/retailcorp-audit, to ingest event streams directly from CloudTrail.

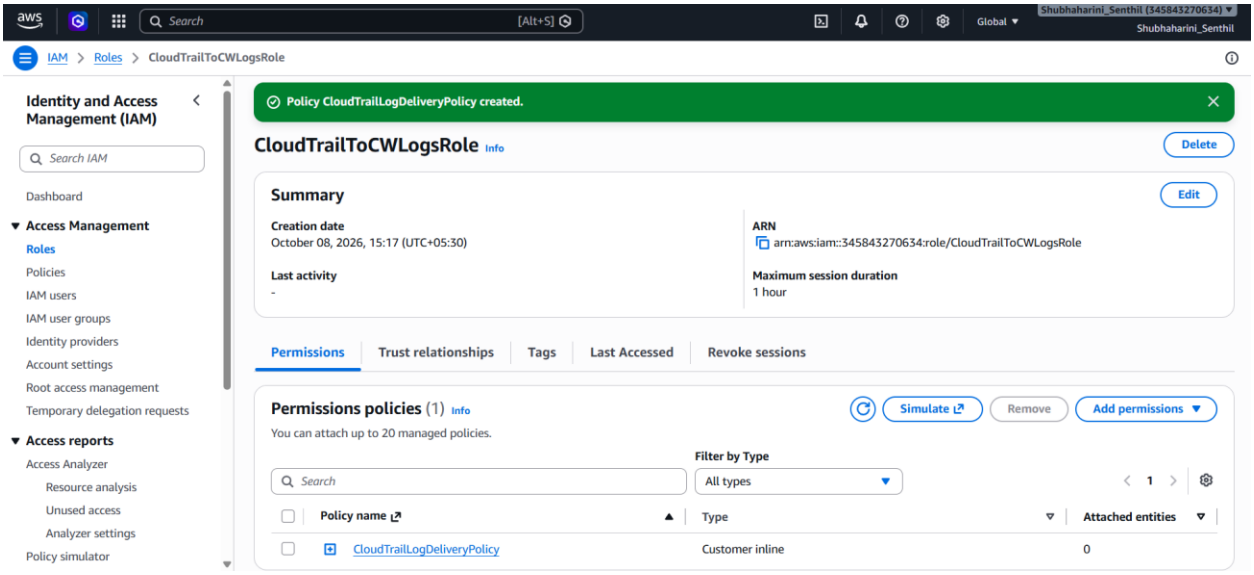


5. i) Configuring IAM Role and Policy for CloudTrail Service Delegation

An IAM service role, CloudTrailToCWLogsRole, was established with a trust policy allowing CloudTrail execution.

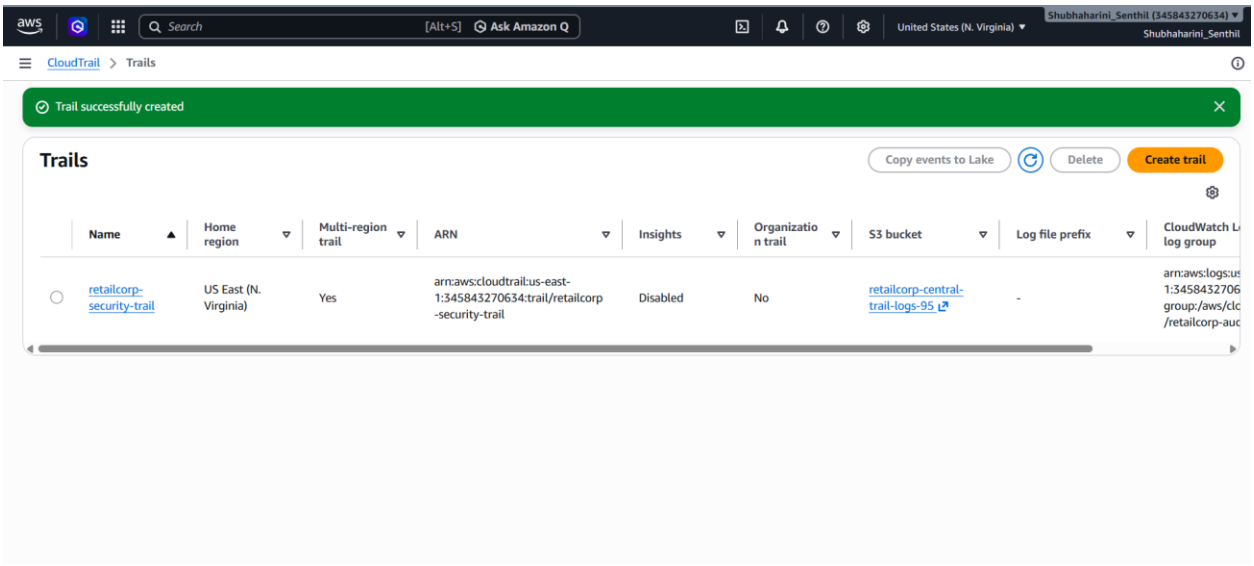


ii) Creating Inline policy for IAM role



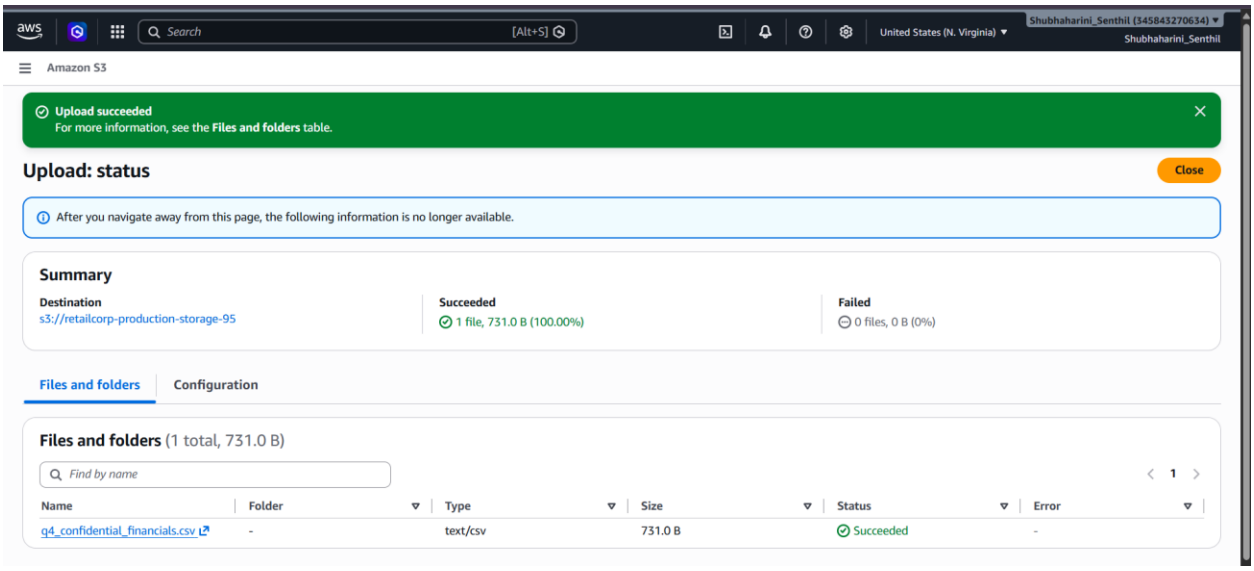
6. Deploying AWS CloudTrail with S3 Object-Level Data Events

Deployed retailcorp-security-trail to execute multi-target telemetry streaming into both S3 and CloudWatch Logs. Unlike default trails that solely record management calls, we enabled object-level Data Events targeting the production bucket, ensuring high-fidelity tracking of granular file operations.



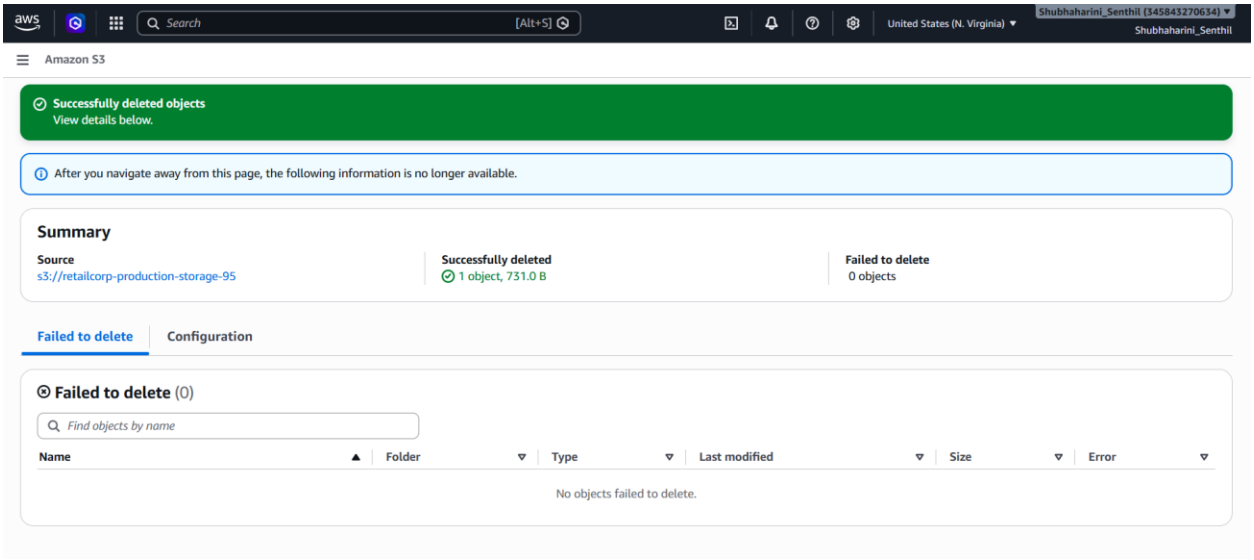
7. Uploading Critical Business Assets to Production Storage

Generated baseline operational activity by uploading an essential business asset, q4_confidential_financials.csv, to the production bucket. This upload registered an initial PutObject data event within CloudTrail, confirming normal data storage workflow before simulating the security incident.



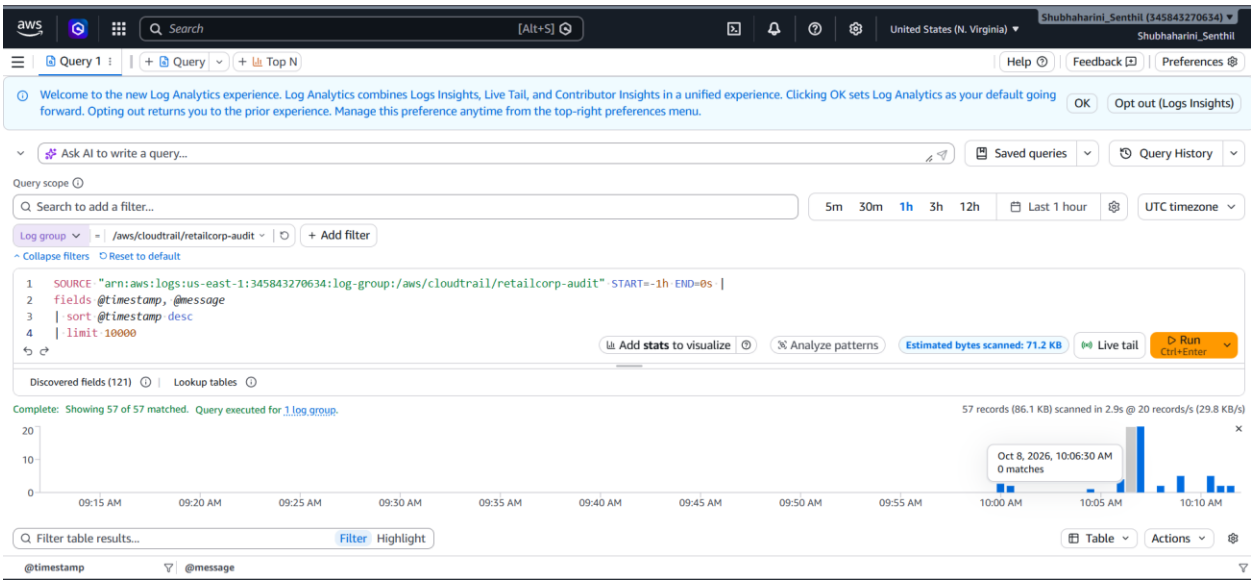
8. Simulating Accidental Deletion Incident

An accidental deletion incident was simulated by an authorized developer identity issuing an unversioned DeleteObject API call against q4_confidential_financials.csv. The object was permanently expunged from the bucket, generating an audit-grade event for the security team to investigate.



9. Executing Forensic Log Investigation in CloudWatch Logs Insights

The incident was investigated using CloudWatch Logs Insights by filtering for DeleteObject events. The query parsed incoming structured JSON streams in seconds, isolating the exact execution timestamp, caller identity ARN, requested bucket key, and originating source IP address.



10. Extracting and Verifying Complete Forensic Audit Record

